AI Prompting Lab Report

1. Introduction

This lab focused on learning how to communicate effectively with AI systems through prompt engineering. The exercises demonstrated how different prompt structures, levels of context, and prompting techniques can significantly influence the quality and accuracy of AI-generated responses.

The lab covered several important concepts including contextual prompting, reasoning prompts, iterative prompting, multimodal AI interaction, AI-assisted application development, data analysis, model comparison, and safe AI usage.

2. What I Learned From the Lab

Throughout the lab, I learned that AI performance depends heavily on the quality of the prompt provided.

Key Learning Areas

A. Context Improves Responses

One of the most important lessons was that providing clear context produces better results than vague instructions. By specifying constraints, audience, and desired output format, the AI generated more relevant and useful responses.

B. Structured Prompting

I learned to organize prompts using:

- Goal
- Context
- Constraints
- Desired Output

This structure helped the AI understand tasks more accurately.

C. Reasoning Prompts

Using phrases such as:

"Think hard before answering"

encouraged the AI to provide deeper analysis and more thoughtful responses when solving complex problems.

D. Iterative Prompting

The lab showed that the first answer is rarely the best answer. Better results were achieved by:

1. Generating multiple options.
2. Providing feedback.
3. Requesting revisions.
4. Expanding the preferred solution.

E. Multimodal AI

I learned that AI can process images and extract information from receipts, handwritten notes, and whiteboards. This extends AI capabilities beyond text-based interactions.

F. AI for Application Development

The lab demonstrated how AI can generate working applications when requirements are clearly specified using Goal, Input, and Output formats.

G. Data Analysis

I learned that AI should be instructed to write and run code when performing calculations. This reduces the risk of incorrect answers and improves transparency.

H. Comparing AI Models

Different AI models often provide different perspectives on the same task. Comparing outputs from multiple models helps identify strengths and weaknesses.

3. How My Prompting Improved During the Process

At the beginning of the lab, my prompts were short and vague.

Example of Initial Prompt

"Help me plan dinner."

The response was general because the AI lacked sufficient information.

Improved Prompt

"I need help planning dinner tonight.

Here is what you need to know:

- I have chicken, rice, onions, and yogurt.
- I only have 30 minutes.
- One person does not eat spicy food.

What I want back:

- Three simple meal options.

- No commentary."

The improved prompt produced significantly better and more relevant suggestions.

As I progressed through the lab, I began adding:

- Clear objectives
- Constraints
- Audience information
- Desired output format

This consistently improved response quality.

4. Prompting Techniques That Helped Me the Most

A. Context-First Prompting

Providing important information before asking the question produced the greatest improvement in AI responses.

Benefits:

- Reduced ambiguity
- More accurate outputs
- Fewer follow-up corrections

B. Think Hard Prompting

Adding:

"Think hard before answering"

resulted in more detailed reasoning and better analysis for complex decisions.

Benefits:

- Better trade-off analysis
- More structured recommendations
- Deeper reasoning

C. Brainstorm–Iterate Method

This was one of the most useful techniques.

Process:

1. Ask for multiple options.

2. Select promising ideas.
3. Provide feedback.
4. Request refinements.

Benefits:

- Higher quality results
- Greater creativity
- Better final solutions

D. Neutral Prompting

I learned that leading questions can bias AI responses.

For example:

Weak:
"Don't you think working from home is clearly better?"

Better:
"Compare working from home versus working in an office. List the strongest arguments for each."

The neutral prompt produced a more balanced and objective answer.

E. Goal–Input–Output Framework

This framework was especially useful when building applications.

Example:

Goal:
Create a countdown timer.

Input:
User presses Start.

Output:
Display a countdown with session alerts.

Benefits:

- Clear requirements
- Better implementation
- Fewer misunderstandings

5. Challenges Faced and Solutions

Challenge 1: Vague Responses

Problem:
Early prompts often generated generic answers.

Solution:
Added detailed context and constraints.

Result:
More personalized and accurate outputs.

Challenge 2: AI Assumptions

Problem:
The AI sometimes assumed missing information.

Solution:
Provided all relevant facts explicitly.

Result:
Reduced misunderstandings.

Challenge 3: Complex Decision-Making

Problem:
Simple prompts did not provide sufficient reasoning.

Solution:
Used "Think hard before answering."

Result:
Received deeper analysis and structured recommendations.

Challenge 4: Choosing the Best Option

Problem:
The first response was not always ideal.

Solution:
Applied the Brainstorm–Iterate Loop.

Result:
Generated improved solutions through multiple revisions.

Challenge 5: Trusting Calculations

Problem:
It was difficult to determine whether AI had actually performed calculations.

Solution:
Requested that the AI write and run code and show the code used.

Result:
Greater confidence in the accuracy of results.